

Sapphire Language Specification and Automation Manual

Current syntax and automation behavior as implemented in the interpreter.

This manual describes the current local implementation. Feature names are not evidence of production scale, hardware availability, or successful external actions.

Core semantics

Sapphire is dynamically typed, uses a tree-walk interpreter, and supports expressions, statements, functions, member access, indexing, and structured process results.

Automation

OS telemetry, notifications, clipboard, filesystem helpers, HTTP helpers, scheduling, and controlled process execution are exposed through standard-library modules.

AI and ML

AI prompt/classification/JSON helpers and local tensor, dataset, model, autodiff, optimizer, and loss utilities are available subject to optional dependencies and backend configuration.

Portability

The source runtime is Python-based. The packaged Windows distribution includes executable tools; source and optional-dependency behavior can differ by environment.

Version 1.0.5. Reproduce local measurements with `benchmarks/benchmark_runtime.py`. Values depend on the machine and environment.